intermediate-elective

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizations and coloring
library(RColorBrewer)
```

Data

```
# vegetation information
veg <- read_csv(here("data", "veg.csv"))

# vegetation meta data information
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

Problems

1. Explain your inspiration

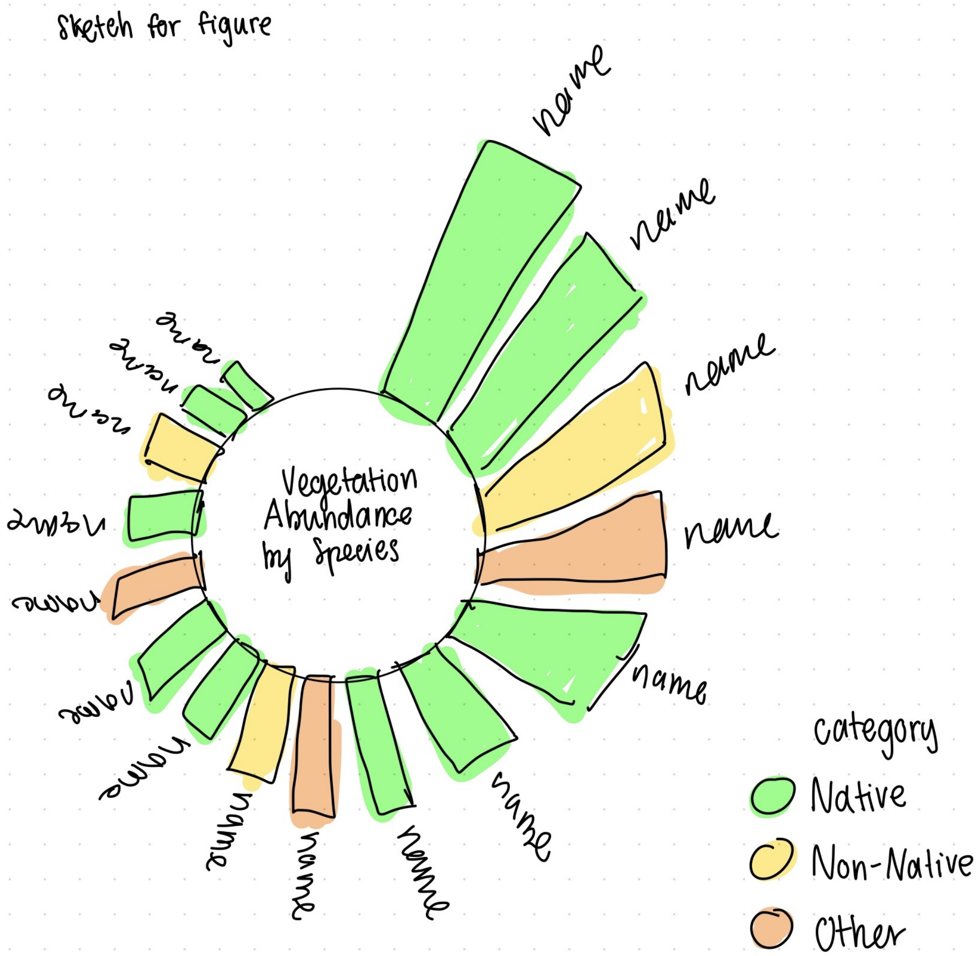
Which visualization you chose: The visualization I chose for inspiration is a circular bar chart made by Cristina Cametti using the IKEA TidyTuesday dataset, which shows the most common IKEA designers grouped by furniture category.

Why it makes sense for your dataset: This works well for my vegetation data because there are many plant species across different cover categories, and the circular layout is a clean way to display a lot of species at once without the labels overlapping like they would in a regular bar chart.

Which variables you're using: The three variables shown are plant species (`psoc`), cover category (`cover_category`), and total percent cover, which is the sum of `percent_cover` across all sampling locations.

How variables map onto visual components: Plant species map onto the individual bars with their names as labels around the outside of the circle, total percent cover maps onto bar height, and cover category maps onto bar color, which mirrors exactly how the IKEA inspiration mapped designer names, item counts, and furniture category onto the same visual components.

2. Plan your figure



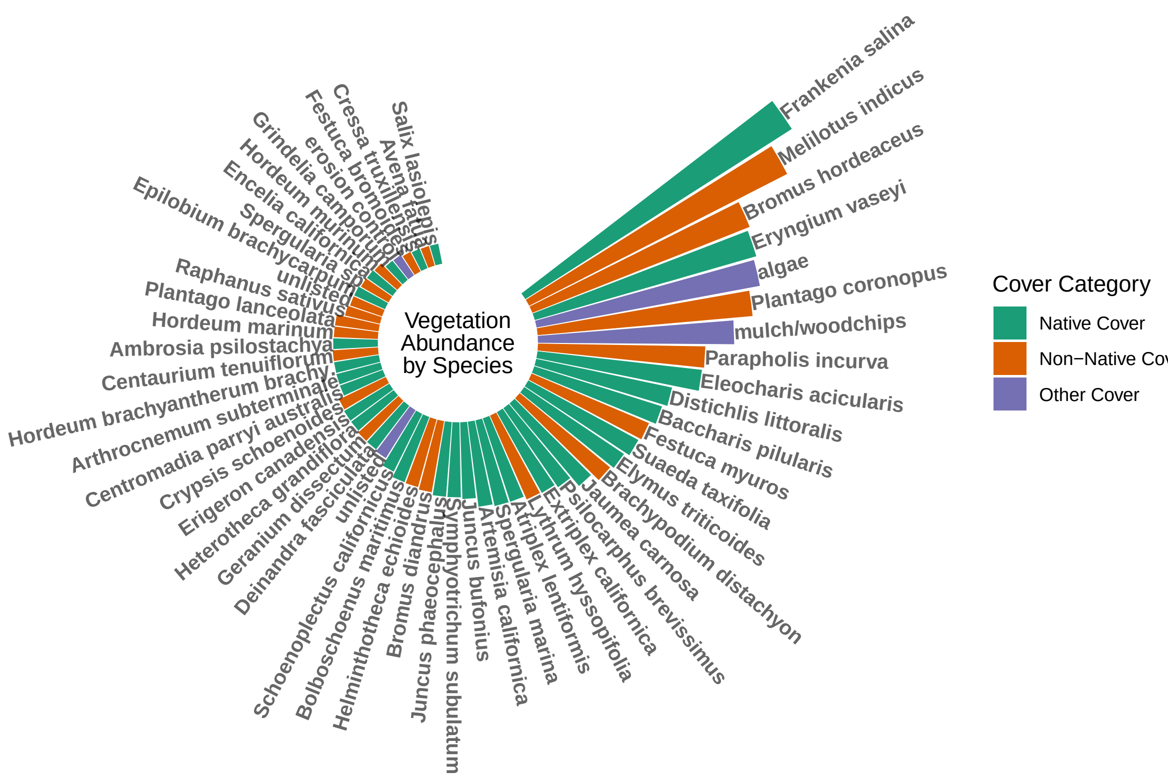
3. Code your figure

```
# wrangling: sum percent cover by species and cover category
# starting with veg data frame
data <- veg |>
  # grouping by cover category and species
  group_by(cover_category, psoc) |>
  # summing percent cover for each group, removing NAs
```

```r
  summarise(n = sum(percent_cover, na.rm = TRUE)) |>
  # ungrouping to avoid grouped operations downstream
  ungroup() |>
  # removing bare ground, thatch, and NA cover categories
  # removing bare ground as it is not a plant species
  filter(!psoc == "Bare Ground") |>
  # removing bare ground cover category
  filter(!cover_category == "BARE GROUND") |>
  # removing thatch
  filter(!cover_category == "THATCH") |>
  # removing literal "NA" entries
  filter(!cover_category == "NA") |>
  # removing true NA values from cover category
  filter(!is.na(cover_category)) |>
  # replacing underscores with spaces in species names
  mutate(psoc = str_replace_all(psoc, "_", " ")) |>
  # renaming cover categories to be more readable
  mutate(cover_category = recode(cover_category,
                                 "NATIVE COVER" = "Native Cover",
                                 "NON-NATIVE COVER" = "Non-Native Cover",
                                 "OTHER COVER" = "Other Cover"))

# keep only species with high abundance to avoid overcrowding visual
dt <- data |>
  # arranging species from highest to lowest summed percent cover
  arrange(desc(n)) |>
  # keeping only species with summed percent cover > 500
  filter(n > 500)

# converting cover category and species to factors
dt <- mutate_at(dt, vars(psoc), as.factor)

# setting factor levels explicitly to exclude NA from legend
dt$cover_category <- factor(dt$cover_category,
                             levels = c("Native Cover", "Non-Native Cover", "Other Cover"))

# adding empty bars at the end for spacing
empty_bar <- 2
# creating a matrix of NAs to use as empty bars
to_add <- matrix(NA, empty_bar, ncol(dt))
# matching column names so rbind works correctly
colnames(to_add) <- colnames(dt)
```

```r
# binding empty bars to bottom of dataset
dt <- rbind(dt, to_add)

# creating an id column to position bars around the circle
dt$id <- seq(1, nrow(dt))

# calculating label angles and alignment for readability
label_data <- dt
# counting total number of bars including empty ones
number_of_bar <- nrow(label_data)
# calculating the angle of each bar's label so text radiates outward
angle <- 90 - 360 * (label_data$id - 0.5) / number_of_bar
# flipping hjust so labels on left side of circle align correctly
label_data$hjust <- ifelse(angle < -90, 1, 0)
# adjusting angle so text on left side of circle is not upside down
label_data$angle <- ifelse(angle < -90, angle + 180, angle)

# setting color palette for cover categories
pal <- brewer.pal(3, "Dark2")

# plotting circular bar chart
plot <- ggplot(dt) +
  # bars filled by cover category
  geom_bar(aes(x = as.factor(id), y = n, fill = cover_category), stat = "identity") +
  # applying color palette, removing NA from legend, and setting legend title
  scale_fill_manual(values = pal,
                    na.translate = FALSE,
                    name = "Cover Category") +
  # setting y limits to create space in the center and top
  ylim(-2000, 8000) +
  # classic theme
  theme_classic() +
  # removing axes and grid lines
  theme(
    axis.text = element_blank(),
    axis.title = element_blank(),
    panel.grid = element_blank(),
    plot.margin = unit(rep(-1, 4), "cm")) +
  # center title
  annotate(geom = "text",
           x = 0, y = -2000,
           hjust = .5, vjust = 0.5,
```

```
        label = "Vegetation\nAbundance\nby Species", # added extra line break
        size = 4,
        lineheight = .8,
        color = "black") +
  # converting to polar/circular layout
  coord_polar(start = 0) +
  # adding species name labels around the outside
  geom_text(data = label_data,
            aes(x = id, y = n + 1, label = psoc, hjust = hjust),
            color = "black", fontface = "bold",
            alpha = 0.6, size = 3.4,
            angle = label_data$angle,
            inherit.aes = FALSE)

# displaying plot
plot
```



4. Describe your process

Process of using someone else's code: To adapt the IKEA circular bar chart code for my vegetation data, I replaced the IKEA dataset with my veg dataset and swapped the relevant variables, replacing `designer` with `psoc` (plant species), `category` with `cover_category`, and replacing `tally()` with `summarise(sum(percent_cover, na.rm = TRUE))` to sum actual percent cover values rather than just counting rows. I also added additional data cleaning steps such as filtering out bare ground, thatch, and NA cover categories, replacing underscores in species names with spaces, and renaming cover categories to be more readable in the legend. Along the way I had to figure out what the code did and why I was using it for the visual.

Challenges: One challenge was that the empty bars added at the end of the dataset for spacing introduced `NA` values into the `cover_category` column, which caused `NA` to appear as its own category in the legend. This was fixed by adding `na.translate = FALSE` inside `scale_fill_manual()`. Another challenge was adjusting the filtering threshold for species abundance, which required running `summary(data$n)` first to understand the distribution before deciding on a cutoff value of 500.

Contrasts with visualizations from class: This visualization differs from those made in class primarily because it uses a polar coordinate system through `coord_polar()`, which transforms a standard bar chart into a circular layout, something we did not use in any class assignments. Unlike the straightforward plots made in class using `geom_point()` with standard x and y axes, this figure requires additional steps like manually calculating bar label angles and adjusting text alignment so that species names radiate outward legibly around the circle.